

EN.553.781: Functional Data Analysis

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0.1 Functional and Shape Data

Definition 0.1.1 (Functional datum). A functional datum is an observation whose basic statistical unit is a function

$$f : D \longrightarrow \mathbb{R}^q,$$

rather than a finite list of unrelated measurements. The domain D is often an interval of time, but may also be a spatial region or a manifold.

A sample in functional data analysis (FDA) is therefore $f_1, \dots, f_n$, where each f_i is viewed as one observation. In practice we observe noisy, discrete samples

$$Y_{ij} = f_i(t_{ij}) + \varepsilon_{ij},$$

possibly at subject-specific times t_{ij} . Smoothing and registration turn these measurements into objects on a common functional domain.

FDA supports familiar statistical tasks:

- means, medians, covariance, and principal modes of variation;
- signal estimation from noisy measurements;
- testing, classification, clustering, and ANOVA;
- regression with functional predictors, responses, or both.

The distinction from multivariate statistics is not merely that the vectors are long. Functions live in infinite-dimensional spaces and carry geometric structure such as smoothness, derivatives, and time transformations.

0.1.1 FDA Versus Multivariate and Time-Series Analysis

If every curve is observed on the same reliable grid $t_1, \dots, t_m$, it can be stored as

$$(f_i(t_1), \dots, f_i(t_m))^T \in \mathbb{R}^m.$$

This representation permits ordinary multivariate tools, but its results may depend on grid density and ignore smoothness. A functional representation allows interpolation, differentiation, integration, irregular sampling, and comparison across grids.

Time-series models typically emphasize sequential dynamics, for example $X_{j+1} = h(X_j) + \varepsilon_j$, with ordered and often equally spaced times. FDA typically treats an entire trajectory as one observation and models cross-sectional variation among trajectories. The two perspectives overlap, but answer different questions.

Example 0.1.1 (Growth curves). For child i , height measurements Y_{ij} are taken at ages t_{ij} . The scientific objects may be the smooth growth curve $f_i(t)$, its velocity $f'_i(t)$, and population variation in the timing of growth spurts. Treating the measurements as an unordered vector loses this derivative and timing structure.

0.1.2 Shape Data and Invariance

Shape analysis is closely related to FDA because curves and surfaces are represented by functions. Following Kendall's principle, shape is what remains after removing transformations that should not change the object, commonly translation, rotation, and scale.

Definition 0.1.2 (Group invariance). Let a group G act on a representation space $\mathcal{F}$. A quantity $S : \mathcal{F} \rightarrow \mathcal{S}$ is invariant to G if

$$S(g \cdot f) = S(f) \quad \text{for every } g \in G.$$

Two objects represent the same shape when they lie in the same orbit $[f] = \{g \cdot f : g \in G\}$.

A shape distance should be defined on equivalence classes, not arbitrary representatives. This motivates quotient spaces later in the course. Typical shape tasks include distances, optimal deformations or geodesics, means, principal modes, and population comparisons.

0.2 Metric and Vector Spaces

Definition 0.2.1 (Metric space). A metric on a set $\mathcal{X}$ is a function $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$ such that for all x, y, z :

1. $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$;
3. $d(x, z) \leq d(x, y) + d(y, z)$.

The pair $(\mathcal{X}, d)$ is a metric space.

Examples include Euclidean distance on $\mathbb{R}^p$ and the uniform distance

$$d_\infty(f, g) = \sup_{t \in [0, 1]} |f(t) - g(t)|$$

on $C([0, 1])$. A metric determines convergence, continuity, neighborhoods, and consequently what it means for two functional observations to be close.

Definition 0.2.2 (Vector space). A real vector space V is a set closed under addition and scalar multiplication satisfying associativity, commutativity of addition, distributivity, a zero element, and additive inverses.

Both $\mathbb{R}^p$ and $C([0, 1])$ are vector spaces under pointwise operations. If $f, g \in C([0, 1])$, then $(af + bg)(t) = af(t) + bg(t)$ is again continuous.

Definition 0.2.3 (Linear independence and basis). Vectors $v_1, \dots, v_m$ are linearly independent if

$$\sum_{j=1}^m a_j v_j = 0 \implies a_1 = \dots = a_m = 0.$$

A basis is a linearly independent spanning set.

Finite-dimensional spaces have finite bases. Function spaces generally require infinite systems, and convergence of an infinite expansion must be specified using a norm.

Definition 0.2.4 (Linear transformation). For vector spaces V, W , a map $T : V \rightarrow W$ is linear when

$$T(au + bv) = aT(u) + bT(v)$$

for every $u, v \in V$ and scalars a, b .

Differentiation, integration, and multiplication by a fixed matrix are linear. The map $f \mapsto f^2$ is not.

0.3 Normed and Banach Spaces

Definition 0.3.1 (Norm). A norm on V is a map $\|\cdot\| : V \rightarrow [0, \infty)$ satisfying

$$\|v\| = 0 \iff v = 0, \quad \|av\| = |a| \|v\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

It induces the metric $d(u, v) = \|u - v\|$.

Important functional norms on $[0, 1]$ are

$$\|f\|_\infty = \sup_t |f(t)|, \quad \|f\|_p = \left(\int_0^1 |f(t)|^p dt \right)^{1/p}.$$

The L^2 norm measures integrated squared size, while the sup norm controls the worst pointwise deviation. Convergence in one need not imply convergence in the other.

Definition 0.3.2 (Cauchy sequence and completeness). A sequence $\{x_n\}$ is Cauchy if for every $\varepsilon > 0$ there is N such that $d(x_m, x_n) < \varepsilon$ whenever $m, n \geq N$. A complete metric space contains the limit of every Cauchy sequence. A complete normed vector space is called a Banach space.

The space $C([0, 1])$ with $\|\cdot\|_\infty$ is Banach. The space $L^p([0, 1])$, formed by identifying functions equal almost everywhere, is Banach for $1 \leq p \leq \infty$.

Remark 0.3.1. The set of continuous functions equipped with the L^2 norm is not complete: an L^2 -Cauchy sequence of continuous functions may converge to a discontinuous equivalence class. Its completion is $L^2([0, 1])$.

0.4 Inner-Product and Hilbert Spaces

Definition 0.4.1 (Inner product). An inner product on a real vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ that is linear in its first argument, symmetric, and positive definite:

$$\langle v, v \rangle \geq 0, \quad \langle v, v \rangle = 0 \iff v = 0.$$

It induces $\|v\| = \sqrt{\langle v, v \rangle}$.

On $\mathbb{R}^p$, $\langle x, y \rangle = x^\top y$. On $L^2([0, 1])$,

$$\langle f, g \rangle_{L^2} = \int_0^1 f(t)g(t) dt. \tag{0.4.1}$$

Theorem 0.4.1 (Cauchy–Schwarz). *For every inner-product space,*

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Corollary 0.4.1 (Triangle inequality). *The norm induced by an inner product satisfies* $\|u + v\| \leq \|u\| + \|v\|$.

Definition 0.4.2 (Hilbert space). A Hilbert space is a complete inner-product space.

The space $L^2([0, 1])$ is the canonical Hilbert space for elementary FDA. Its geometry supports angles, orthogonality, projections, and least squares just as Euclidean space does.

0.4.1 Orthogonality and Projection

Elements u, v are orthogonal, written $u \perp v$, if $\langle u, v \rangle = 0$. An orthonormal set $\{e_j\}$ satisfies $\langle e_j, e_k \rangle = \mathbb{1}\{j = k\}$.

Theorem 0.4.2 (Projection theorem). *Let M be a closed linear subspace of a Hilbert space H . For every $f \in H$ there is a unique $m^* \in M$ minimizing $\|f - m\|$. It is characterized by*

$$f - m^* \perp M.$$

For $M = \text{span}(e_1, \dots, e_J)$ with an orthonormal basis,

$$P_M f = \sum_{j=1}^J \langle f, e_j \rangle e_j.$$

This is the infinite-dimensional origin of the normal equations in least squares.

0.4.2 Orthonormal Bases and Fourier Expansion

Definition 0.4.3 (Complete orthonormal system). An orthonormal sequence $\{e_j\}_{j \geq 1}$ is complete if the only element orthogonal to every e_j is zero. Equivalently, its linear span is dense in the Hilbert space.

Theorem 0.4.3 (Hilbert-space expansion). *For a complete orthonormal system and $f \in H$,*

$$f = \sum_{j=1}^{\infty} \langle f, e_j \rangle e_j \quad \text{in } H, \quad \|f\|^2 = \sum_{j=1}^{\infty} |\langle f, e_j \rangle|^2.$$

The second identity is Parseval's identity. On $L^2([0, 1])$, a real Fourier system consists of the constant function and appropriately normalized $\sin(2\pi kt)$ and $\cos(2\pi kt)$. Fourier bases are natural for periodic functions; splines are usually more local and flexible near boundaries.

Example 0.4.1 (Truncation). The approximation

$$f_J(t) = \sum_{j=1}^J c_j e_j(t), \quad c_j = \langle f, e_j \rangle,$$

is the best J -term approximation within $\text{span}(e_1, \dots, e_J)$ under L^2 loss. Smooth functions typically have rapidly decaying coefficients in a compatible basis.

0.5 From Discrete Measurements to Smooth Functions

For one subject, suppose (t_i, y_i) satisfy $y_i = f(t_i) + \varepsilon_i$. Choose basis functions $b_1, \dots, b_J$ and approximate

$$f(t) = \sum_{j=1}^J c_j b_j(t).$$

Let $B_{ij} = b_j(t_i)$, $c = (c_1, \dots, c_J)^\top$, and $y = (y_1, \dots, y_m)^\top$. Least-squares fitting becomes

$$\hat{c} = \arg \min_c \|y - Bc\|^2. \quad (0.5.1)$$

If B has full column rank, $\hat{c} = (B^\top B)^{-1} B^\top y$. In computation one should use a QR or SVD solve rather than explicitly form the inverse.

The number and type of basis functions control the bias–variance tradeoff. Too few cannot represent the signal; too many follow measurement noise. A common coefficient vector has meaning only when all curves use the same basis and domain.

0.5.1 Roughness-Penalized Estimation

Instead of choosing a very small basis, one may directly penalize roughness:

$$\hat{f} = \arg \min_f \left\{ \sum_{i=1}^m \{y_i - f(t_i)\}^2 + \lambda \int_0^1 \{f^{(r)}(t)\}^2 dt \right\}. \quad (0.5.2)$$

Usually $r = 2$, so curvature is penalized. The tuning parameter λ controls the tradeoff:

$$\lambda \downarrow 0 : \text{flexible fit}, \quad \lambda \uparrow \infty : \text{approach the null space of } D^r.$$

For $r = 2$, the null space consists of linear functions.

With $f(t) = b(t)^\top c$, define

$$R_{jk} = \int_0^1 b_j^{(r)}(t) b_k^{(r)}(t) dt.$$

Then

$$\hat{c}_\lambda = (B^\top B + \lambda R)^{-1} B^\top y. \quad (0.5.3)$$

This resembles ridge regression, but R penalizes rough directions rather than all coefficient directions equally.

Proposition 0.5.1 (Linear-smoother form). *At the observation times, the penalized fitted values are*

$$\hat{y} = S_\lambda y, \quad S_\lambda = B(B^\top B + \lambda R)^{-1} B^\top.$$

Their effective degrees of freedom are $\text{df}(\lambda) = \text{tr}(S_\lambda)$.

Cross-validation or generalized cross-validation can select λ . With irregular or sparse subject-specific sampling, smoothing choices can strongly affect later covariance estimates and should be propagated or at least reported.

Theorem 0.5.1 (Smoothing-spline representation). *For squared error with the integrated squared second-derivative penalty, the minimizer over the appropriate Sobolev space is a natural cubic spline with knots at the distinct observation times.*

Thus an infinite-dimensional optimization reduces to a finite-dimensional one without first imposing an arbitrary dense knot grid.

0.6 Statistical Summaries of Functional Data

Let F be a random element of $L^2([0, 1])$ with $\mathbb{E} \|F\|^2 < \infty$. The population mean function is

$$\mu(t) = \mathbb{E}[F(t)],$$

where the pointwise expression is interpreted when evaluation is meaningful. As an element of L^2 , the sample mean is

$$\hat{\mu}(t) = \frac{1}{n} \sum_{i=1}^n f_i(t).$$

Proposition 0.6.1 (Mean as a Fréchet mean). *The mean $\mu = \mathbb{E}F$ uniquely minimizes*

$$g \mapsto \mathbb{E} \|F - g\|_{L^2}^2.$$

Similarly, $\hat{\mu}$ minimizes $n^{-1} \sum_i \|f_i - g\|_{L^2}^2$.

Proof. The Hilbert-space identity

$$\mathbb{E} \|F - g\|^2 = \mathbb{E} \|F - \mu\|^2 + \|\mu - g\|^2$$

follows because $\mathbb{E}(F - \mu) = 0$ makes the cross term vanish. □

This variational definition generalizes to nonlinear spaces: replace the Hilbert norm by an appropriate metric and minimize the sum of squared distances. Uniqueness is no longer automatic.

0.6.1 Covariance Function

Definition 0.6.1 (Covariance function). For a square-integrable process F ,

$$K(s, t) = \text{Cov}\{F(s), F(t)\} = \mathbb{E}[(F(s) - \mu(s))(F(t) - \mu(t))].$$

The covariance is symmetric and nonnegative definite:

$$K(s, t) = K(t, s), \quad \sum_{i,j=1}^m a_i a_j K(t_i, t_j) \geq 0.$$

Its empirical estimate from densely observed curves is

$$\widehat{K}(s, t) = \frac{1}{n-1} \sum_{i=1}^n \{f_i(s) - \hat{\mu}(s)\} \{f_i(t) - \hat{\mu}(t)\}.$$

The diagonal $K(t, t)$ is pointwise variance, while off-diagonal values describe association across locations. Unlike a finite covariance matrix, K is a function on a two-dimensional domain.

0.7 Linear Operators on Function Spaces

Definition 0.7.1 (Bounded linear operator). For normed spaces V, W , a linear map $T : V \rightarrow W$ is bounded if there is $C < \infty$ such that

$$\|Tf\|_W \leq C \|f\|_V \quad \text{for every } f \in V.$$

Its operator norm is $\|T\|_{\text{op}} = \sup_{\|f\| \leq 1} \|Tf\|$.

For linear maps between normed spaces, boundedness is equivalent to continuity. In finite dimensions every linear map is bounded; this is false in infinite dimensions. For example, differentiation is not bounded from $C^1([0, 1])$ equipped only with the sup norm into $C([0, 1])$.

0.7.1 Integral Operators

Given a kernel $k : [0, 1]^2 \rightarrow \mathbb{R}$, define

$$(Tf)(t) = \int_0^1 k(s, t) f(s) \, ds. \quad (0.7.1)$$

This map is linear. If $k \in L^2([0, 1]^2)$, then T is a bounded Hilbert–Schmidt operator on L^2 and

$$\|T\|_{\text{op}} \leq \left\{ \int_0^1 \int_0^1 k(s, t)^2 \, ds \, dt \right\}^{1/2}.$$

Definition 0.7.2 (Adjoint). For Hilbert spaces H_1, H_2 and bounded $T : H_1 \rightarrow H_2$, the adjoint $T^* : H_2 \rightarrow H_1$ is the unique operator satisfying

$$\langle Tf, g \rangle_{H_2} = \langle f, T^*g \rangle_{H_1}$$

for all f, g .

For matrices, the adjoint is the transpose. For the integral operator (0.7.1),

$$(T^*g)(s) = \int_0^1 k(s, t) g(t) \, dt.$$

Thus T is self-adjoint when $k(s, t) = k(t, s)$ almost everywhere.

Definition 0.7.3 (Positive operator). A self-adjoint operator T is positive semidefinite if $\langle Tf, f \rangle \geq 0$ for every f .

0.7.2 Covariance Operator

The covariance function induces

$$(\mathcal{K}f)(t) = \int_0^1 K(s, t) f(s) \, ds. \quad (0.7.2)$$

Equivalently, in Hilbert notation,

$$\mathcal{K}f = \mathbb{E}[\langle F - \mu, f \rangle (F - \mu)].$$

Proposition 0.7.1. *The covariance operator is self-adjoint and positive semidefinite. If $\mathbb{E} \|F - \mu\|^2 < \infty$, it is trace class and*

$$\mathrm{tr}(\mathcal{K}) = \mathbb{E} \|F - \mu\|^2 = \int_0^1 K(t, t) \, dt.$$

For any f ,

$$\langle \mathcal{K}f, f \rangle = \mathbb{E}[\langle F - \mu, f \rangle^2] \geq 0.$$

This is the functional analogue of $a^\top \Sigma a \geq 0$.

0.8 Principal Component Analysis

For centered vectors $X \in \mathbb{R}^p$ with covariance Σ , the first principal direction solves

$$v_1 = \arg \max_{\|v\|=1} \mathrm{Var}(v^\top X) = \arg \max_{\|v\|=1} v^\top \Sigma v.$$

Theorem 0.8.1 (Rayleigh quotient). *For symmetric Σ , the maximum of $v^\top \Sigma v$ over $\|v\| = 1$ is the largest eigenvalue λ_1 , attained by a corresponding eigenvector. Subsequent principal directions are eigenvectors obtained under orthogonality constraints.*

For observations $x_1, \dots, x_n$, the rank- q PCA subspace also minimizes reconstruction error:

$$\min_{U^\top U = I_q} \sum_{i=1}^n \|x_i - UU^\top x_i\|^2.$$

If the centered data matrix has SVD $X = UDV^\top$, loadings are columns of V , scores are columns of UD , and explained variances are proportional to d_j^2 .

0.9 Functional Principal Component Analysis

Functional PCA (FPCA) replaces the covariance matrix by the covariance operator. A principal direction ϕ is a unit-norm function maximizing

$$\mathrm{Var}\{\langle F - \mu, \phi \rangle\} = \langle \mathcal{K}\phi, \phi \rangle.$$

The solution satisfies the eigenequation

$$\mathcal{K}\phi_j = \lambda_j \phi_j, \quad \int_0^1 K(s, t) \phi_j(s) \, ds = \lambda_j \phi_j(t). \quad (0.9.1)$$

Theorem 0.9.1 (Spectral decomposition of covariance). *Let $\mathcal{K}$ be a compact, self-adjoint, positive covariance operator. Then it has orthonormal eigenfunctions $\{\phi_j\}$ with nonnegative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \rightarrow 0$, and*

$$K(s, t) = \sum_{j=1}^{\infty} \lambda_j \phi_j(s) \phi_j(t)$$

in the appropriate L^2 sense.

0.9.1 Karhunen–Loève Expansion

Define the principal component scores

$$\xi_{ij} = \langle f_i - \mu, \phi_j \rangle = \int_0^1 \{f_i(t) - \mu(t)\} \phi_j(t) dt.$$

Theorem 0.9.2 (Karhunen–Loève). *Under the preceding second-moment assumptions,*

$$F(t) = \mu(t) + \sum_{j=1}^{\infty} \xi_j \phi_j(t)$$

in mean square, where

$$\mathbb{E}\xi_j = 0, \quad \text{Cov}(\xi_j, \xi_k) = \lambda_j \mathbb{1}\{j = k\}.$$

Truncation gives the model

$$f_i(t) \approx \mu(t) + \sum_{j=1}^q \xi_{ij} \phi_j(t). \quad (0.9.2)$$

The fraction of integrated variance explained by the first q components is

$$\frac{\sum_{j=1}^q \lambda_j}{\sum_{j=1}^{\infty} \lambda_j}.$$

Proposition 0.9.1 (Optimal reconstruction). *Among all q -dimensional linear subspaces of L^2 , the span of $\phi_1, \dots, \phi_q$ minimizes expected squared reconstruction error, which equals $\sum_{j>q} \lambda_j$.*

0.9.2 Computation from a Common Grid

Suppose every centered curve is sampled on a common grid $t_1, \dots, t_m$. Stack curve vectors as rows of $X_c \in \mathbb{R}^{n \times m}$. Ordinary PCA of X_c approximates FPCA only after accounting for numerical integration. With quadrature weights $W = \text{diag}(w_1, \dots, w_m)$, the discrete inner product is

$$\langle f, g \rangle \approx f^\top W g.$$

An SVD of $X_c W^{1/2}$ gives eigenfunctions normalized in the weighted inner product. Ignoring unequal grid spacing changes the geometry.

Alternatively, if $f_i(t) = b(t)^\top c_i$ and $G = \int b(t)b(t)^\top dt$, then functional orthonormality is measured by G , not the Euclidean coefficient norm. FPCA becomes a generalized eigenproblem or an ordinary eigenproblem after whitening by $G^{1/2}$.

Example 0.9.1 (Interpreting a mode). Plot

$$\hat{\mu}(t) \pm c\sqrt{\hat{\lambda}_j} \hat{\phi}_j(t)$$

for a small constant c , often one or two. These curves visualize the functional change associated with positive and negative scores. The sign of an eigenfunction is arbitrary, so interpretation must be invariant to a simultaneous sign change of ϕ_j and its scores.

0.9.3 Practical Qualifications

- Measurement error inflates the empirical covariance diagonal. Smooth the covariance surface or use a model that separates signal and noise rather than treating the diagonal spike as functional variation.
- Dense common grids permit direct covariance estimation. Sparse, irregular designs require pooled smoothing and conditional score estimation.
- Eigenfunctions with nearly tied eigenvalues are individually unstable; their joint eigenspace can still be stable.
- Standard FPCA measures amplitude variation under L^2 . Timing variation can dominate the components unless curves are registered.